Electronic Engineering and
Communications Technology Department

Semester: Fall
Exam Date: 8 /12/ 2020

QUESTIONS FOR THE MID TERM EXAM

Subject: (ELC 422) Digital Signal Processing
Examiners: Dr. Samir M. Kamal & Dr. Sara M. Hassan
Spec.: Level- 4, Elec. Eng. & Comm.Tech. Dep.
Time: 60 Min

Number of Pages: one | Number of Questions: 2 | Attempt all questions

Model - A

Question 1 [20 Points]

a- In A/D, in detail, explain the sampling process of a continuous time signals.

b- Check the periodicity of the following signals: $X(n) = \cos(3n\pi)$

c- For the signal $x(t) = 5\cos(100\pi t)$ to be sampled with sampling frequency F_S = 300Hz.

(i) Find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P. Calculate the samples values of the first cycle in x(n) then plot them at the same figure with x(t).

Question 2 [20 Points]

a- Briefly, Classify the different signal types.

b- Prove that the DFT. X(K), of a sequence x(n) is periodic.

c- For the following sequence: x(n) = {1, 0, 1, 1}:

(i) Using the DFT algorithm find the frequency components of x(n) and check your results by calculating x(n) using the IFFT method.

(ii) Using your results, verify Parseval's theorem.

(iii) Plot the amplitude and the phase spectrums.

(iv) Since we use FFT algorithm instead of DFT algorithm, what is its computational saving?

GOOD LUCK